

Adobe - 100 Technical MCQ Questions

Adobe (Software Engineer, New Grad)

For BTech Computer Science / IT Students

Object-Oriented Programming

- 1. Which access modifier makes a member accessible only within the same class?**
 - A) public
 - B) protected
 - C) private
 - D) default
- 2. Constructor overloading is possible when:**
 - A) Constructors have same name but different parameter lists
 - B) Constructors are private
 - C) A class has no constructor
 - D) Only one constructor exists
- 3. Which concept allows objects of different classes to be treated as objects of a common superclass?**
 - A) Polymorphism
 - B) Encapsulation
 - C) Abstraction
 - D) Overloading
- 4. Which OOP principle is primarily used to reduce code duplication by reusing existing classes?**
 - A) Inheritance
 - B) Polymorphism
 - C) Abstraction
 - D) Encapsulation
- 5. What is a virtual function used for in C++?**
 - A) To achieve runtime polymorphism
 - B) To avoid inheritance
 - C) To create static memory
 - D) To improve compilation speed
- 6. What is the purpose of the 'super' keyword in Java?**
 - A) To create a static method
 - B) To refer to the parent class members
 - C) To delete an object
 - D) To define an interface

7. A class that cannot be instantiated directly is called a(n):

- A) Abstract class
- B) Final class
- C) Static class
- D) Sealed class

8. Which of the following is true about a constructor?

- A) It has a return type
- B) It is called automatically when an object is created
- C) It can be inherited like other methods
- D) It must be private

9. Which of these is an example of compile-time polymorphism?

- A) Method overriding
- B) Operator overloading (in C++)
- C) Dynamic method dispatch
- D) Runtime type identification

10. What happens if a class does not implement all abstract methods of an interface it implements?

- A) Compilation succeeds silently
- B) The class must be declared abstract
- C) Runtime error occurs
- D) Nothing happens

DBMS & SQL

11. Which normal form removes partial dependency on a composite primary key?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

12. What does ACID stand for in DBMS transactions?

- A) Atomicity, Consistency, Isolation, Durability
- B) Accuracy, Consistency, Integration, Data
- C) Atomic, Complete, Independent, Durable
- D) Association, Consistency, Isolation, Data

13. Which property of a transaction ensures that partial updates are never visible to other transactions?

- A) Atomicity
- B) Consistency
- C) Isolation
- D) Durability

14. What is a primary key?

- A) A column that can contain duplicate values
- B) A column (or set of columns) that uniquely identifies each row and cannot be NULL
- C) An index used only for sorting
- D) A key used only in foreign tables

15. What does DBMS stand for?

- A) Database Management System
- B) Data Backup Management Service
- C) Database Model Structure
- D) Data Base Modeling Software

16. A foreign key is used to:

- A) Uniquely identify a row in its own table
- B) Establish a link between two tables
- C) Encrypt sensitive data
- D) Speed up all queries automatically

17. What is a deadlock in the context of a database transaction?

- A) Two or more transactions waiting indefinitely for resources locked by each other
- B) A syntax error in SQL
- C) A missing index
- D) A type of join

18. Which type of JOIN returns all rows from the left table and matched rows from the right table?

- A) INNER JOIN
- B) LEFT JOIN
- C) RIGHT JOIN
- D) CROSS JOIN

19. Which of the following commands removes all rows from a table but keeps its structure, and cannot be rolled back in most databases?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) REMOVE

20. Which SQL clause is used to filter groups after a GROUP BY?

- A) WHERE
- B) HAVING
- C) FILTER
- D) ORDER BY

Operating Systems

21. Which of these is a necessary condition to prevent deadlock by breaking 'circular wait'?

- A) Impose a strict ordering of resource acquisition
- B) Allow unlimited resource requests
- C) Never release resources
- D) Increase the number of processes

22. Which of the following best describes multiprogramming?

- A) Running multiple programs on multiple CPUs simultaneously only
- B) Keeping multiple programs in memory so the CPU always has one to execute, improving utilization
- C) Running only one program at a time
- D) A type of scheduling exclusive to real-time systems

23. What is spooling?

- A) Simultaneous Peripheral Operations On-Line - buffering data for devices like printers
- B) A type of deadlock
- C) A page replacement algorithm
- D) A CPU scheduling technique

24. What is the banker's algorithm used for?

- A) Deadlock avoidance by checking safe states before allocating resources
- B) Memory compression
- C) File compression
- D) CPU scheduling

25. What does the term 'fragmentation' refer to in memory management?

- A) Wasted memory space due to allocation and deallocation patterns (internal or external)
- B) A type of process priority
- C) A CPU scheduling delay
- D) A file system backup issue

26. Which of the following is true about a microkernel architecture?

- A) It moves most services to user space, keeping the kernel minimal
- B) It has no scheduler
- C) It cannot support IPC
- D) It is always slower with no exceptions

27. What is the purpose of a context switch?

- A) To save the state of a running process and load the state of another
- B) To terminate a process
- C) To allocate memory
- D) To format a disk

28. What is a critical section in OS?

- A) A section of code that accesses shared resources and must not be executed by more than one process/thread at a time
- B) A section reserved for the kernel only
- C) A type of page fault
- D) A memory segment for stack only

29. Which scheduling algorithm gives the shortest average waiting time (in theory, for a given batch)?

- A) FCFS
- B) SJF (Shortest Job First)
- C) Round Robin with a large quantum
- D) Priority Scheduling

30. What is a monolithic kernel?

- A) A kernel where all OS services run in kernel space as a single large process
- B) A kernel with no drivers
- C) A kernel used only in embedded systems
- D) A kernel that cannot be modified

Computer Networks

31. Which protocol is used for secure remote login to a server?

- A) Telnet
- B) SSH
- C) FTP
- D) SMTP

32. What is the function of a DNS server?

- A) To translate human-readable domain names into IP addresses
- B) To assign private IP addresses within a LAN
- C) To encrypt HTTP traffic
- D) To manage email delivery

33. What is a MAC address?

- A) A unique hardware identifier assigned to a network interface
- B) An IP address used only for routing
- C) A type of DNS record
- D) A session identifier

34. What is the primary purpose of a load balancer?

- A) To distribute incoming network traffic across multiple servers to improve reliability and performance
- B) To encrypt data at rest
- C) To act as a firewall only
- D) To assign static IP addresses

Data Structures & Algorithms

35. Which data structure uses FIFO (First In First Out) order?

- A) Stack
- B) Queue
- C) Tree
- D) Graph

- 36. Which data structure uses LIFO (Last In First Out) order?**
- A) Queue
 - B) Stack
 - C) Linked List
 - D) Array
- 37. Which data structure is typically used to implement BFS?**
- A) Stack
 - B) Queue
 - C) Heap
 - D) Array only
- 38. Which sorting algorithm is considered stable and has $O(n \log n)$ time complexity in all cases?**
- A) QuickSort
 - B) Merge Sort
 - C) Selection Sort
 - D) Heap Sort
- 39. What is the time complexity of finding the minimum element in an unsorted array?**
- A) $O(1)$
 - B) $O(\log n)$
 - C) $O(n)$
 - D) $O(n \log n)$
- 40. Which algorithmic technique does Merge Sort primarily use?**
- A) Greedy
 - B) Divide and Conquer
 - C) Dynamic Programming
 - D) Backtracking
- 41. Which data structure is best suited for implementing a priority queue efficiently?**
- A) Array
 - B) Linked List
 - C) Heap
 - D) Stack
- 42. What is the time complexity of accessing an element in an array by index?**
- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n \log n)$
- 43. What is the time complexity of building a heap from an unsorted array (heapify)?**
- A) $O(n \log n)$
 - B) $O(n)$
 - C) $O(\log n)$
 - D) $O(n^2)$

44. What is the worst-case time complexity of QuickSort?

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n)$

45. Which of the following best describes a trie data structure?

- A) A tree-like structure used for efficient retrieval of strings, especially for prefix matching
- B) A type of balanced binary search tree
- C) A hashing algorithm
- D) A graph traversal method

46. Which of the following problems is a classic example solved using dynamic programming?

- A) Binary search
- B) 0/1 Knapsack problem
- C) Bubble sort
- D) Linear search

47. Which data structure is typically used to implement DFS (iteratively)?

- A) Queue
- B) Stack
- C) Heap
- D) Hash table

48. What is the time complexity of binary search on a sorted array of n elements?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

49. What is the time complexity of inserting an element at the head of a singly linked list?

- A) $O(n)$
- B) $O(1)$
- C) $O(\log n)$
- D) $O(n \log n)$

50. Which algorithm is used to find the Minimum Spanning Tree of a graph?

- A) Dijkstra's Algorithm
- B) Kruskal's or Prim's Algorithm
- C) Bellman-Ford Algorithm
- D) Floyd-Warshall Algorithm

51. What is the average time complexity of searching in a well-implemented hash table?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n^2)$

52. Which of the following is a common technique to resolve hash collisions?

- A) Chaining or open addressing
- B) Binary search
- C) Recursion
- D) Memoization

53. Which data structure underlies the implementation of recursion internally?

- A) Queue
- B) Stack (call stack)
- C) Heap (memory heap)
- D) Graph

54. What is dynamic programming primarily used for?

- A) Solving problems by breaking them into overlapping subproblems and storing results to avoid recomputation
- B) Sorting arrays only
- C) Random number generation
- D) Compiling code faster

55. What does BFS stand for in graph traversal?

- A) Best First Search
- B) Breadth First Search
- C) Binary First Search
- D) Backtrack First Search

56. What is the primary difference between a greedy algorithm and dynamic programming?

- A) Greedy makes a locally optimal choice at each step without reconsidering, DP considers overlapping subproblems and often guarantees a globally optimal solution
- B) They are the same technique
- C) DP never uses recursion
- D) Greedy always gives an optimal solution while DP does not

57. What is the key property of a min-heap?

- A) The parent node is always greater than its children
- B) The parent node is always smaller than or equal to its children
- C) It must be a complete binary tree only, with no ordering
- D) It is always a balanced BST

58. What is the time complexity of Bubble Sort in the worst case?

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(n^2)$
- D) $O(\log n)$

59. What is the space complexity of an iterative approach to computing Fibonacci numbers with memoization storing all n results?

- A) $O(1)$
- B) $O(n)$
- C) $O(n^2)$
- D) $O(\log n)$

60. What is the amortized time complexity of appending an element to a dynamic array (like Python's list or Java's ArrayList)?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$ amortized
- D) $O(n^2)$

61. What is memoization?

- A) Storing the results of expensive function calls and reusing them when the same inputs occur again
- B) A sorting technique
- C) A type of recursion without a base case
- D) A method to delete duplicate array elements

62. What is a hash collision?

- A) When two different keys map to the same hash bucket/index
- B) When a hash table runs out of memory
- C) When a key is deleted incorrectly
- D) When two hash tables merge

63. Which data structure would you use to check if a string of brackets is balanced?

- A) Queue
- B) Stack
- C) Array
- D) Linked List

64. Which technique explores all possible options and backtracks when a solution path fails, often used in puzzles like N-Queens?

- A) Greedy algorithm
- B) Backtracking
- C) Dynamic programming
- D) Binary search

65. What is a balanced binary search tree used to prevent?

- A) Overflow errors
- B) Degeneration into a linked list (worst-case $O(n)$ operations)
- C) Memory leaks
- D) Integer overflow

66. Which graph algorithm is used to find the shortest path in a graph with possible negative edge weights (no negative cycles)?

- A) Dijkstra's Algorithm
- B) Bellman-Ford Algorithm
- C) Prim's Algorithm
- D) Kruskal's Algorithm

67. Which traversal of a Binary Search Tree gives elements in sorted order?

- A) Pre-order
- B) Post-order
- C) In-order
- D) Level-order

68. What is the time complexity of Dijkstra's algorithm using a binary heap?

- A) $O(V^2)$
- B) $O(E \log V)$
- C) $O(V + E)$
- D) $O(E^2)$

69. What is a monotonic stack typically used for?

- A) Finding the next greater/smaller element efficiently
- B) Sorting an array in place
- C) Implementing recursion
- D) Hashing

70. What is the time complexity of the Two-Pointer technique on a sorted array to find a pair with a given sum?

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

Programming Language Fundamentals

71. In Java, what is the difference between '==' and '.equals()' when comparing objects?

- A) They always behave identically
- B) '==' compares references while '.equals()' compares content (if overridden)
- C) '.equals()' is only for primitives
- D) '==' is only used for Strings

72. In Python, which of the following is a mutable data type?

- A) Tuple
- B) String
- C) List
- D) int

73. Which of the following best describes garbage collection?

- A) A manual process where the developer frees memory explicitly
- B) An automatic process that reclaims memory occupied by objects no longer in use
- C) A compilation step
- D) A type of exception handling

74. In C++, what does the 'const' keyword do when applied to a variable?

- A) Makes the variable a pointer
- B) Prevents the variable's value from being modified after initialization
- C) Declares a static variable
- D) Declares a global variable

75. Which keyword is used to handle exceptions in Python?

- A) catch
- B) except
- C) rescue
- D) handle

76. In C, what does the 'malloc' function do?

- A) Deallocates memory
- B) Allocates a block of memory dynamically
- C) Initializes a variable to zero
- D) Declares a static variable

77. What does 'pass by reference' mean?

- A) A copy of the variable's value is passed to the function
- B) The function receives a reference to the actual variable, so changes affect the original
- C) Variables cannot be passed to functions
- D) Only primitive types can be passed

78. What is method chaining?

- A) Calling multiple methods on the same object in a single statement by having each method return the object itself
- B) A way to declare multiple classes
- C) A type of exception handling
- D) A memory management technique

79. What is a memory leak?

- A) When allocated memory is never freed even though it is no longer needed, wasting resources over time
- B) When a program uses too little memory
- C) A type of compilation error
- D) A CPU scheduling issue

80. What is a lambda function/expression?

- A) A named function stored in a file
- B) An anonymous, inline function typically used for short operations
- C) A type of class
- D) A compiler directive

81. Which of the following is a value type (not a reference type) in Java?

- A) int
- B) String
- C) Array
- D) Object

82. What is the default value of an uninitialized boolean instance variable in Java?

- A) true
- B) false
- C) null
- D) 0

83. What is the purpose of the 'finally' block in exception handling (Java/Python)?

- A) It executes only if an exception occurs
- B) It executes only if no exception occurs
- C) It always executes regardless of whether an exception occurred or not
- D) It is used to declare exceptions

84. What is the difference between a compiler and an interpreter?

- A) A compiler translates the whole program to machine code before execution, an interpreter executes code line-by-line
- B) They are identical
- C) Interpreters are always faster than compilers
- D) Compilers cannot detect syntax errors

Web & Cloud Fundamentals

85. What is the purpose of cookies in web applications?

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

86. What is a RESTful API?

- A) An API that strictly requires SOAP protocol
- B) An architectural style for designing networked applications using stateless operations over standard HTTP methods
- C) A type of database
- D) A programming language

87. What does HTTP stand for?

- A) HyperText Transfer Protocol
- B) HighText Transmission Protocol
- C) Hyperlink Transfer Text Protocol
- D) HyperText Trade Protocol

88. What is auto-scaling in cloud computing?

- A) Automatically adjusting compute resources up or down based on demand
- B) Manually adding servers only
- C) A type of database indexing
- D) A billing model only

89. What is the difference between IaaS, PaaS, and SaaS?

- A) They are all the same cloud service model
- B) IaaS provides raw infrastructure, PaaS provides a platform to build apps, SaaS delivers ready-to-use software
- C) SaaS provides infrastructure while IaaS provides software
- D) PaaS is only for storage

90. What does CSS stand for?

- A) Cascading Style Sheets
- B) Computer Style Sheets
- C) Creative Style System
- D) Cascading Software Sheets

91. What is the primary difference between HTTP and HTTPS?

- A) HTTPS uses encryption (SSL/TLS) to secure data in transit while HTTP does not
- B) HTTP is faster in all cases
- C) HTTPS cannot use port 443
- D) There is no difference

92. Which HTTP method is typically used to retrieve data without side effects?

- A) POST
- B) GET
- C) DELETE
- D) PUT

System Design Basics

93. What is the purpose of a message queue (e.g., Kafka, RabbitMQ) in system design?

- A) To decouple producers and consumers of data, enabling asynchronous processing and buffering
- B) To replace a database
- C) To render a webpage
- D) To manage user sessions only

94. What is a load balancer used for in system design?

- A) To store data persistently
- B) To distribute incoming requests across multiple servers to avoid overload on any single server
- C) To compress images
- D) To manage user authentication only

95. What is eventual consistency?

- A) A guarantee that all replicas are updated instantly
- B) A consistency model where, given enough time without new updates, all replicas converge to the same value
- C) A model where data is never consistent
- D) A synonym for strong consistency

96. What is the CAP theorem in distributed systems?

- A) A system can simultaneously guarantee Consistency, Availability, and Partition tolerance
- B) A distributed system can only guarantee two of Consistency, Availability, and Partition tolerance at the same time during a network partition
- C) A theorem about compiler design
- D) A theorem unrelated to distributed systems

97. What is caching used for in system design?

- A) To permanently store all application data
- B) To temporarily store frequently accessed data closer to the application for faster retrieval
- C) To replace a database entirely
- D) To handle user authentication

98. What is database sharding used for?

- A) To horizontally partition a large database across multiple servers to improve scalability
- B) To encrypt a database
- C) To back up data
- D) To normalize tables

99. What is horizontal scaling?

- A) Adding more power (CPU/RAM) to an existing server
- B) Adding more servers/machines to distribute the load
- C) Reducing the number of servers
- D) Only applicable to databases

100. What is vertical scaling?

- A) Adding more machines to a system
- B) Increasing the resources (CPU/RAM) of an existing machine
- C) Distributing data across regions
- D) A type of caching

Answer Key

1-C	2-A	3-A	4-A	5-A	6-B	7-A	8-B	9-B	10-B
11-B	12-A	13-C	14-B	15-A	16-B	17-A	18-B	19-C	20-B
21-A	22-B	23-A	24-A	25-A	26-A	27-A	28-A	29-B	30-A
31-B	32-A	33-A	34-A	35-B	36-B	37-B	38-B	39-C	40-B
41-C	42-C	43-B	44-B	45-A	46-B	47-B	48-B	49-B	50-B
51-C	52-A	53-B	54-A	55-B	56-A	57-B	58-C	59-B	60-C
61-A	62-A	63-B	64-B	65-B	66-B	67-C	68-B	69-A	70-C
71-B	72-C	73-B	74-B	75-B	76-B	77-B	78-A	79-A	80-B
81-A	82-B	83-C	84-A	85-A	86-B	87-A	88-A	89-B	90-A
91-A	92-B	93-A	94-B	95-B	96-B	97-B	98-A	99-B	100-B

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.